MA 5104 Statistical Inference and Learning **due: 21-Oct-2026, 12 PM**
HW–5 Dept. of Mathematics, IIT Madras

Topic: Statistical Learning

Important note for submissions:

- Only handwritten answers are allowed.
- The hard-copy of solutions should be submitted to me in Monday lecture.
- No email submissions will be entertained.
- **Submit solutions to ALL Problems. Give BRIEF explanations. No more than two both-sided A4-sized page should be submitted.**
- ONLY submit solutions to Problems 3, 5, 9, 10, 13.

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1. Monthly rent (Rs. thousand) is regressed on floor area (sq ft) and whether the flat is furnished. Reference = unfurnished.

$$\widehat{\text{rent}} = 6.0 + 0.035\,\text{area} + 4.5 \cdot \mathbf{1}\{\text{furnished}\} + 0.012\,\text{area} \cdot \mathbf{1}\{\text{furnished}\}$$

   A property agent reports "furnishing adds Rs. 4,500 a month." Is this correct? State what 4.5 actually measures, and give the correct rent for an 800 sq ft furnished flat.

2. The same model is refitted with *furnished* as the reference category instead of *unfurnished*. Which of the four coefficients change, and which fitted values change?

3. Crop yield (quintals/hectare) on fertiliser applied (kg/hectare), with and without a soil-quality index in the model:

	intercept	fertiliser	soil quality
Model 1	18.40	**+0.62**	—
Model 2	11.20	**−0.15**	3.80

   Is one of the two fits wrong? Explain what each fertiliser coefficient is estimating, and offer a reason why farmers' behaviour could produce this pattern.

4. `sales ~ advertising_spend + channel` is fitted with no interaction term, in a market where an extra rupee of advertising does far more for the online channel than for the retail channel. What does the single fitted advertising slope estimate, and for whom is it wrong?

5. A soil-type predictor has three levels: clay, loam, sand. Explain why it enters the design matrix as two columns rather than three, and state what goes wrong if you use three.

6. A trial records mean yield (quintals/hectare) for two seed varieties at two irrigation levels:

	low water	high water
Variety A	32	48
Variety B	44	38

The state agriculture board asks you for "the effect of switching to high irrigation" as a single number for its advisory. Answer the board.

7. A damage-severity rating with levels *minor*, *moderate*, *severe* is coded as $1, 2, 3$ and entered as one column. Name the two assumptions this silently imposes.

8. In a dataset of shipments, one predictor is parcel weight in grams and another is declared value in lakhs of rupees. State one quantity that is completely unaffected by this choice of units, and explain why gradient descent *is* affected.

9. Standardising the predictors changes nothing about an ordinary least squares fit, but changes the answer given by ridge. Explain the difference, one sentence each.

10. A student fits kNN with $k = 1$ to a dataset of 500 loan applications and reports zero training error. What has been demonstrated? The same student now refits with $k = n = 500$. Describe the bias and the variance of the prediction at a point x_0.

11. A vendor demonstrates a demand-forecasting model and quotes its error on the data used to build it as evidence of forecasting accuracy. Name the missing step, and say in which direction the quoted number is wrong.

12. A lab has expression measurements on 2,000 genes for 40 patients and fits least squares to predict a clinical score. The fit passes exactly through all 40 observations. Give two separate reasons why this is bad news rather than good.

13. Collinearity leaves fitted values stable while making individual coefficients unstable. Give one setting in which this would not trouble you, and one in which it would be fatal.

14. An analyst standardises every predictor using the mean and standard deviation of the **full** dataset, and then runs 10-fold cross-validation. State the general rule being broken, and say whether the resulting estimate comes out too large or too small.

15. Two-fold and ten-fold cross-validation are run on the same data with the same method. Which reports the larger error, and why?